

Solve One-Step Addition and Subtraction Equations

Lesson 7-2

Name: _____

Date: _____

Class: _____

Key Vocabulary

Level 2 Standard

Picture first, then the word, then a plain-language meaning. Say each word out loud.

$$x + 2 = 7$$

balanced with =

$x + 5 = 12$ — both sides equal 12 when $x = 7$

Equation

Write the definition:

$$x + 2 = 7$$

balanced with =

Addition undoes subtraction: if $x - 3 = 10$, add 3 to
get $x = 13$

Inverse operation

Write the definition:

$$x + 2 = 7$$

balanced with =

$x + 5 = 12 \rightarrow$ subtract 5 from both sides $\rightarrow x = 7$ (x is
isolated)

Isolate

Write the definition:

$$x + 2 = 7$$

balanced with =

$x = 7$ is the solution to $x + 5 = 12$ because $7 + 5 = 12$
✓

Solution

Write the definition:



number

$x + 4 = 9$ is solved in one step: subtract 4 from both sides, so $x = 5$.

One-step equation

Write the definition:



number

If you subtract 4 from the left side, you must subtract 4 from the right to keep balance.

Balance

Write the definition:

Guided Notes

Level 2 Standard



WHAT WE'RE LEARNING TODAY

I can solve one-step addition and subtraction equations using inverse operations.



Fill in each blank as we go. Use the Word Bank to help you.



WORD BANK — FILL EACH BLANK WITH THE BEST WORD

Equation

Inverse operation

Isolate

Solution

One-step equation

Balance

1 A math sentence with an equal sign showing two equal amounts is an _____.

2 An operation that undoes another, like subtraction undoing addition, is an _____.

3 To get the variable alone on one side of the equation is to _____ it.

4 The value of the variable that makes the equation true is the _____.

5 An equation solved in a single step is a _____.

6 Doing the same thing to both sides to keep them equal keeps the equation in _____.



Watch & Try — Worked Examples

See the notes in action: watch one worked all the way through, then try the next with the same steps.

 I do — watch

Follow each step as your teacher solves it.

Problem: Solve: $x + 9 = 15$

- A. $x = 6$
- B. $x = 24$
- C. $x = 9$
- D. $x = 15$

Step 1 Subtract 9 from both sides: $x = 15 - 9 = 6$.

Step 2 Check: $6 + 9 = 15$ ✓

✓ **Answer:** A. $x = 6$

 We do — together

Solve this one with your class using the same steps.

Problem: Solve: $y - 7 = 20$

- A. $y = 27$
- B. $y = 13$
- C. $y = 7$
- D. $y = 140$

Step 1 _____

Step 2 _____

Answer: _____

 You do — your turn

Now try one on your own. Show every step.

Problem: Solve: $n + 15 = 42$

- A. $n = 27$
- B. $n = 57$
- C. $n = 15$
- D. $n = 42$

Show your work:

Try It

Solve on your own. Check the answer key when you are done.

1. Unlock Card 2: Solve $n - 4 = 9$.

- A. $n = 13$
- B. $n = 5$
- C. $n = 36$
- D. $n = -5$

Show your work:

2. Unlock Card 3: A vault held some clue tokens. After the detective added 16 tokens, there were 50. The equation is $t + 16 = 50$. How many tokens were there before?

- A. 34
- B. 66
- C. 16
- D. 800

Show your work:

Stretch Your Thinking

Level 2 enrichment

Challenge task — explain your reasoning in full sentences.

A detective solved $x + 25 = 60$ and got $x = 35$. Another detective solved $y - 25 = 60$ and got $y = 35$. Are both correct? Explain using inverse operations and check each answer.

Sentence starter: For $x + 25 = 60$, the inverse operation is ____, so $x =$ ____ . Check: ____ . For $y - 25 = 60$, the inverse operation is ____, so $y =$ ____ . Check: ____ . Therefore, ____ .

Show your work:

Reflect — Exit Ticket

Solve: $p + 2.8 = 9.1$

- A. $p = 6.3$
- B. $p = 11.9$
- C. $p = 6.8$
- D. $p = 3.25$

Your answer:

Answer Key & Teacher Guide

1. **Notes 1:** Equation
2. **Notes 2:** Inverse operation
3. **Notes 3:** Isolate
4. **Notes 4:** Solution
5. **Notes 5:** One-step equation
6. **Notes 6:** Balance
7. **We Try (notes):** A. $y = 27$ — *Add 7 to both sides: $y = 20 + 7 = 27$. Check: $27 - 7 = 20$ ✓*
8. **You Try (notes):** A. $n = 27$ — *Subtract 15 from both sides: $n = 42 - 15 = 27$. Check: $27 + 15 = 42$ ✓*
9. **Try It 1:** A. $n = 13$ — *Add 4 to both sides: $n = 9 + 4 = 13$. Check: $13 - 4 = 9$ ✓*
10. **Try It 2:** A. 34 — *Subtract 16 from both sides: $t = 50 - 16 = 34$. Check: $34 + 16 = 50$ ✓*
11. **Exit Ticket:** A. $p = 6.3$ — *Subtract 2.8 from both sides: $p = 9.1 - 2.8 = 6.3$. Check: $6.3 + 2.8 = 9.1$ ✓*